

Convex Functions

Two Characterizations and Closure Properties

Theorems 22.2 and 22.3

Convex Optimization Problems

Where we are going today

- **Warm-up:** convex sets, and the *epigraph* of a function.

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The big idea

Convexity of a *function* = convexity of a *set* (its epigraph).
Everything today is that one sentence, unpacked.

Definition 1

A set $S \subseteq \mathbb{R}^m$ is **convex** if for all $p, q \in S$ and all $\alpha \in (0, 1)$,

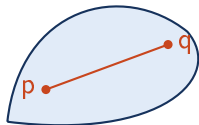
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Prerequisite 1: Convex sets

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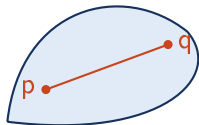
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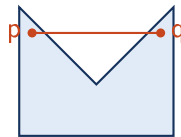
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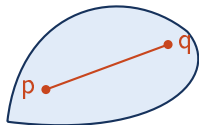
Not convex: some chord leaves the set.

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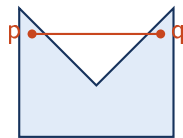
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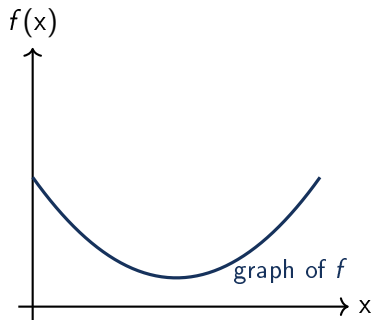
$\alpha p + (1 - \alpha)q$ traces the segment from q ($\alpha = 0$) to p ($\alpha = 1$).

Prerequisite 2: Graph, epigraph, and convex functions

Let $\Omega \subseteq \mathbb{R}^n$ be convex and $f : \Omega \rightarrow \mathbb{R}$.

Graph

$$\{[x^\top, f(x)]^\top : x \in \Omega\} \subset \mathbb{R}^{n+1}$$



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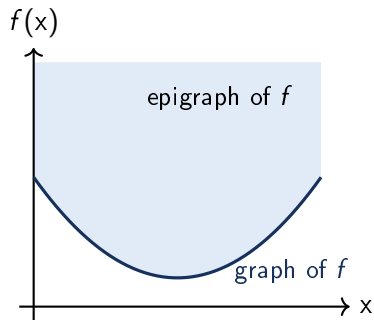
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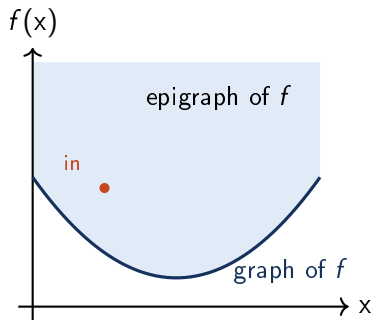
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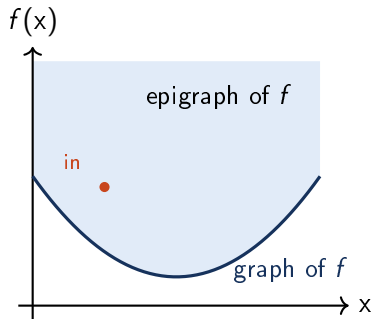
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Epigraph

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Definition (convex function)

f is convex $\iff$ $\text{epi}(f)$ is a convex subset of $\mathbb{R}^{n+1}$.



A point $[x^\top, \beta]^\top$ is in $\text{epi}(f)$ exactly when the height β is at least $f(x)$.

Theorem 22.2: the “chord above the graph” test

Theorem 2 (22.2)

Let $\Omega \subseteq \mathbb{R}^n$ be convex. A function $f : \Omega \rightarrow \mathbb{R}$ is convex if and only if for all $x, y \in \Omega$ and all $\alpha \in (0, 1)$,

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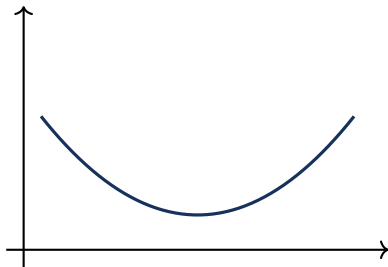
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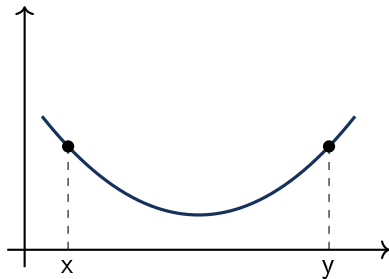
Why care?

- Our definition talks about a set in $\mathbb{R}^{n+1}$: awkward to check.
- This inequality involves only f evaluated at three points of Ω : easy to check.
- Many textbooks take this inequality as *the* definition.

Reading the inequality geometrically (one variable, $n = 1$)

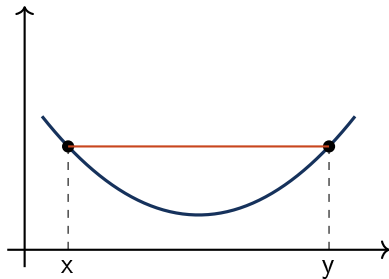


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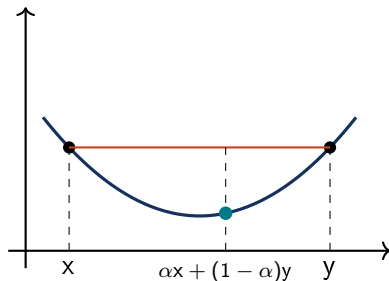
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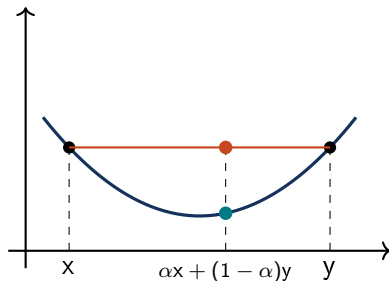
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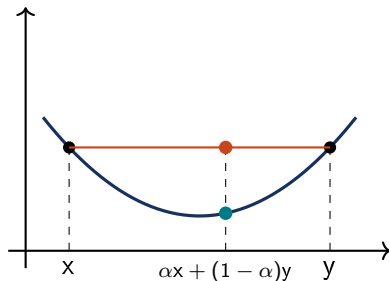
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- **Orange:** $\alpha f(x) + (1 - \alpha)f(y)$, height of the chord.
- **Convex** $\iff$ teal $\leq$ orange, always.

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Common thread

In both directions we use the *same* dictionary:

$$[x^\top, a]^\top \in \text{epi}(f) \iff x \in \Omega \text{ and } f(x) \leq a.$$

Proof of 22.2, ($\Leftarrow$): Step 1, set up

Assume: $f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$ for all $x, y \in \Omega$, $\alpha \in (0, 1)$.

Goal: $\text{epi}(f)$ is convex.

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Proof of 22.2, ($\Leftarrow$): Step 2, chain the inequalities

We know: (H) $f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$, (E) $f(x) \leq a$, $f(y) \leq b$.

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Result

$$f(\alpha x + (1 - \alpha)y) \leq \alpha a + (1 - \alpha)b.$$

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Pause & think

Where did we use that Ω is convex?

Proof of 22.2, ($\Rightarrow$): Step 1, choose good points

Assume: f is convex, i.e. $\text{epi}(f) \subset \mathbb{R}^{n+1}$ is a convex set.

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Why choose $a = f(x)$ and $b = f(y)$ rather than larger numbers? (Hint: we want the *sharpest* bound at the end.)

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Proof of 22.2, ($\Rightarrow$): Step 3, substitute back

Recall $a = f(x)$ and $b = f(y)$. Then

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Summary of Theorem 22.2

$\text{epi}(f)$ convex $\iff$ chord lies on or above the graph.

Same dictionary both ways. The ($\Leftarrow$) direction used $f(x) \leq a$; the ($\Rightarrow$) direction chose $a = f(x)$.

Theorem 3 (22.3)

Let $f, f_1, f_2 : \Omega \rightarrow \mathbb{R}$ be convex functions on a convex set Ω . Then

- (a) for every $c \geq 0$, the function cf is convex;
- (b) $f_1 + f_2$ is convex.

Theorem 22.3: building new convex functions

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Plan: no epigraphs needed. We use Theorem 22.2 as the working definition and just manipulate inequalities.

Theorem 22.3: building new convex functions

Theorem 3 (22.3)

Let $f, f_1, f_2 : \Omega \rightarrow \mathbb{R}$ be convex functions on a convex set Ω . Then

- (a) for every $c \geq 0$, the function cf is convex;*
- (b) $f_1 + f_2$ is convex.*

Plan: no epigraphs needed. We use Theorem 22.2 as the working definition and just manipulate inequalities.

Notation: the book writes $a \geq 0$ for the scalar; we use c to avoid clashing with the height a from before.

Proof of 22.3(a): scaling

Fix $x, y \in \Omega$, $\alpha \in (0, 1)$, $c \geq 0$. Let $\tilde{f} = cf$.

$$\tilde{f}(\alpha x + (1 - \alpha)y) = c f(\alpha x + (1 - \alpha)y)$$

Proof of 22.3(a): scaling

Fix $x, y \in \Omega$, $\alpha \in (0, 1)$, $c \geq 0$. Let $\tilde{f} = cf$.

$$\tilde{f}(\alpha x + (1 - \alpha)y) = c f(\alpha x + (1 - \alpha)y) \quad (\text{definition of } \tilde{f})$$

Proof of 22.3(a): scaling

Fix $x, y \in \Omega$, $\alpha \in (0, 1)$, $c \geq 0$. Let $\tilde{f} = cf$.

$$\begin{aligned}\tilde{f}(\alpha x + (1 - \alpha)y) &= c f(\alpha x + (1 - \alpha)y) && \text{(definition of } \tilde{f}) \\ &\leq c(\alpha f(x) + (1 - \alpha)f(y)) && (f \text{ convex, and } c \geq 0)\end{aligned}$$

Proof of 22.3(a): scaling

Fix $x, y \in \Omega$, $\alpha \in (0, 1)$, $c \geq 0$. Let $\tilde{f} = cf$.

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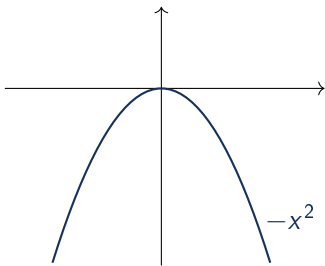
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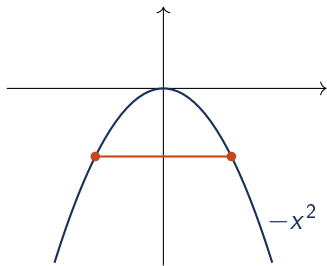
Pause & think

Where exactly did $c \geq 0$ matter? Try $f(x) = x^2$ and $c = -1$: is $-x^2$ convex?

Why $c \geq 0$ is needed: a quick check

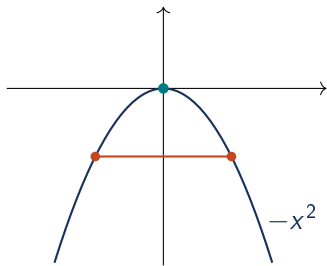


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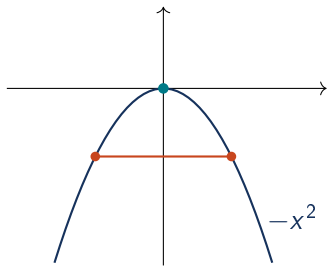
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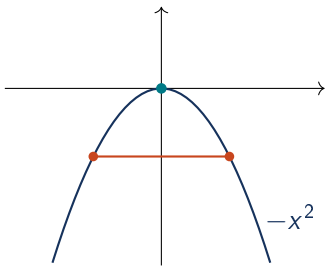
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- But the graph at the midpoint is at height $0 > -1$.
- The chord is **below** the graph, so $-x^2$ is *not* convex.
- Multiplying by a negative number **flips** $\leq$.

Let $f_3 = f_1 + f_2$. Fix $x, y \in \Omega$ and $\alpha \in (0, 1)$.

$$f_3(\alpha x + (1 - \alpha)y) = f_1(\alpha x + (1 - \alpha)y) + f_2(\alpha x + (1 - \alpha)y)$$

Proof of 22.3(b): sums

Let $f_3 = f_1 + f_2$. Fix $x, y \in \Omega$ and $\alpha \in (0, 1)$.

$$\begin{aligned} f_3(\alpha x + (1 - \alpha)y) &= f_1(\alpha x + (1 - \alpha)y) + f_2(\alpha x + (1 - \alpha)y) \\ &\leq (\alpha f_1(x) + (1 - \alpha)f_1(y)) + (\alpha f_2(x) + (1 - \alpha)f_2(y)) \\ &\quad \text{by convexity of } f_1 \text{ and of } f_2 \text{ (add two inequalities)} \end{aligned}$$

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By Theorem 22.2, $f_3 = f_1 + f_2$ is convex. ■

Corollary (nonnegative combinations)

If $f_1, \dots, f_\ell$ are convex and $c_1, \dots, c_\ell \geq 0$, then $c_1 f_1 + \dots + c_\ell f_\ell$ is convex.

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Definition 4 (22.4: strict convexity)

$f : \Omega \rightarrow \mathbb{R}$ is **strictly convex** if for all $x \neq y$ in Ω and $\alpha \in (0, 1)$,

$$f(\alpha x + (1 - \alpha)y) < \alpha f(x) + (1 - \alpha)f(y).$$

Consequences and a definition

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Strict convexity forbids the chord from touching the graph between x and y .

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Take-home summary

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3. **Thm 22.3:** nonnegative multiples and sums of convex functions are convex.
4. **Technique:** to prove convexity, pick x, y, α , then manipulate inequalities **until you reach the chord inequality**.

Try at home

- Show $f(x) = |x|$ is convex. (Use the triangle inequality.)
- Show $\max\{f_1, f_2\}$ is convex.
- Show that a strictly convex function cannot be constant on any segment.